

Sprint 6 Report (Dates: Apr 3, 2026 – Apr 10, 2026)

YouTube link of Sprint 6 Video: <https://www.youtube.com/watch?v=rmxwIzKjaLA>

What's New (User Facing)

- Persistent control mode + clearer toggle UX: Joining a world no longer forces PLAYER vs SERVER_BOT; the last selected mode is preserved and the Ctrl+M switch message is clearer.
- More reliable action execution: Discrete actions (use/select_slot/attack) are prioritized so they don't time out behind long movement/look queues; discrete action timeout defaults were increased (2s → 5s).
- Better learning stability + long-run soak support: Observation handling was fixed to avoid dropping intermediate states; policy shutdown/checkpointing is safer on disconnect; soak-test tooling was added to support longer runs.
- Remote / distributed bridge support + packaging: Bridge URI can be configured for VPS/remote runs; first-run runtime overlay defaults are created automatically; repo layout + Windows/Linux packaging scripts/docs were improved.

Work Summary (Developer Facing)

This sprint focused on turning recent AI-agent work into something stable for real runs and easier to deploy. We improved the Forge↔Bridge control path (persisted control mode, clearer feedback, prioritized discrete steps) and hardened the Python bridge + DQN learning loop against timing/queue issues (no longer draining observations each tick, avoiding half-open transitions that corrupt learning, and ensuring clean shutdown/checkpoint behavior on disconnect). In parallel, we expanded distributed/remote readiness (role-safe hello ordering, runtime overlay defaults, bridge URI resolver) and invested heavily in packaging and layout refactors so the project works in both dev and packaged environments, including Linux build flows. The main barriers were coordination across a large refactor (directory moves, runtime paths) and platform-specific packaging constraints; we addressed them with runtime path centralization, build scripts, and targeted tests to prevent regressions.

Unfinished Work

N/A

Completed Issues/User Stories

Here are links to the issues that we completed in this sprint:

Issue 1:

https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/bridge/server.py

Issue 2:

https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/bridge/policy_worker.py

Issue 3:

https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/ai_rl/dqn/agent.py

Issue 4:

https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/ai_rl/dqn/agent.py

Reminders (Remove this section when you save the file):

- * Each issue should be assigned to a milestone
- * Each completed issue should be assigned to a pull request
- * Each completed pull request should include a link to a "Before and After" video
- * All team members who contributed to the issue should be assigned to it on GitHub
- * Each issue should be assigned story points using a label
- * Story points contribution of each team member should be indicated in a comment

Incomplete Issues/User Stories

Here are links to issues we worked on but did not complete in this sprint:

N/A

Code Files for Review

Please review the following code files, which were actively developed during this sprint, for quality:

[server.py]

(https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/bridge/server.py)

[policy_worker.py]

(https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/bridge/policy_worker.py)

[agent.py]

(https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/ai/rl/dqn/agent.py)

[ClientBridgeHooks.java]

(https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint_6/ai_agent_project/bot/forged/src/main/java/com/example/aiagent/client/ClientBridgeHooks.java)

Retrospective Summary

Here's what went well:

- Strong technical stabilization: fixed action-queue timing issues and learning poisoning risks that directly impact correctness.
- Deployment progress: runtime paths + overlay defaults + Linux packaging made the project much closer to “runs anywhere.”
- Testing improved confidence: added/expanded tests for connection lifecycle, control mode, overlay defaults, and DQN behavior.

Here's what we'd like to improve:

- Earlier issue/PR hygiene: create GitHub issues/PR links as we go so sprint reporting is faster.
- Better timeboxing for refactors: large directory/packaging shifts can balloon; we should gate them behind clearer milestones.
- Documentation parallelization: keep Notion/file notes moving while code refactors land.

Here are changes we plan to implement in the next sprint:

- Finish remaining Notion/file notes in priority order (DQN → Java server bridge → movement/tick logic).
- Run longer soak + remote bridge validation and capture a short “before/after” demo clip.
- Tighten release workflow (scripts/docs) so Windows + Linux builds are repeatable without manual fixes.